

An Open, Browser-Native, Mechanistically Explicit Lung Simulator for Mechanical Ventilation Education

Architecture, computational verification, and mobile deployment

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Central contribution: The novelty of this work is not the existence of web-based ventilator simulation, which has a published history spanning more than two decades. It is the integration of an inspectable three-compartment mechanistic lung model, explicit recruitment state, strict phenotype/scenario separation, direction-aware failure semantics, solver instrumentation, reproducible acceptance testing, and responsive browser deployment in a single open educational artifact.

Keywords: mechanical ventilation; simulation-based education; respiratory mechanics; recruitment; browser-based simulation; computational verification; open-source software; numerical methods

Abstract

Background. Web-based mechanical ventilation simulators have demonstrated educational value, including non-inferior learning outcomes compared with conventional on-site lung simulators. However, published reviews and expert commentary have also highlighted marked heterogeneity among software simulators, limited standards for validation, and infrequent publication of underlying mathematical models. In parallel, patient-specific virtual-patient and digital-twin research has advanced rapidly, but these systems address a different problem: individualized physiologic prediction rather than transparent, low-barrier education.

Objective. To describe and computationally verify an open, browser-native mechanistic lung simulator designed to occupy the middle ground between simplified educational interfaces and patient-specific research digital twins. The design goal was to make both physiology and numerical behavior inspectable: learners and reviewers can see the model assumptions, alter ventilator settings, examine compartment behavior, and distinguish structural infeasibility from numerical non-convergence.

Methods. The simulator represents the lung as three quasi-static tissue compartments governed by finite-capacity exponential elastic laws, central and branch airway resistance, recruitment state, and airway opening pressure. Phenotype parameters are separated from scenario-specific decisions, including initial PEEP, recruitment state, ventilator mode, and ventilator settings. The mechanics solver uses an analytic Jacobian, scaled convergence criteria, direction-aware feasibility checks, explicit rollback on failed steps, and machine-readable per-step solver instrumentation. Version 0.4.5 adds a responsive mobile/desktop interface, volume-control and pressure-control workflows, browser Web Worker execution, cancellation support, inline errors, JSON export, and validation of settings that would eliminate expiratory time. The release was evaluated with a deterministic automated acceptance suite.

Results. The v0.4.5 release passed 143 of 143 automated assertions with zero failures. The test battery covers initialization contracts, analytic Jacobian agreement, small-signal time-constant behavior, flow and pressure conservation, low-resistance behavior, recruitment and closed-compartment invariants, multi-breath stability, timestep convergence, failure-state rollback, positive- and negative-flow boundary classification, controller/tracker rollback, scenario validation, and solver instrumentation. The underlying constitutive mechanics, recruitment equations, phenotype parameters, and gas-exchange module are unchanged from the preceding verified core release; the principal v0.4.5 additions are deployment, interaction, and failure-handling improvements.

Conclusions. This work contributes a reproducible, inspectable software architecture for mechanical ventilation education rather than a claim of patient-specific clinical prediction. Its novelty is the combination of mechanistic transparency, explicit numerical failure semantics, automated computational verification, and browser/mobile accessibility. The simulator is an educational exploration prototype, not a clinical decision tool and not a patient-specific digital twin. Physiologic validation, expert face validity, and educational-outcome studies remain necessary.

1. Introduction

Mechanical ventilation is conceptually difficult because learners must integrate respiratory physiology, ventilator control logic, waveform interpretation, and dynamic consequences of parameter changes. Simulation is therefore attractive: it allows repeated experimentation without patient risk and can expose learners to scenarios that may be uncommon or unsafe to reproduce at the bedside. Mechanical ventilation simulation has been used in physical test lungs, high-fidelity mannequin systems, and software-based environments, and recent evidence supports the educational utility of virtual simulation.

Web-based simulation itself is not new. Takeuchi and colleagues described an Internet-based virtual ventilator in 2004 that combined a respiratory model, spontaneous-breath model, and ventilator module. A later comparative review identified multiple web-accessible mechanical ventilation virtual simulators and documented substantial heterogeneity in respiratory mechanics, gas exchange, modes, monitoring, and usability. More recently, VentSim has been evaluated in critical care fellows and was non-inferior to an on-site simulator for teaching ventilator waveform analysis. These studies establish that web-based ventilation simulation is feasible and educationally relevant.

The unresolved problem is different: transparency and verifiability. Contemporary commentary on mechanical ventilation simulation notes that there are no established validation standards for software simulators and that few publish their underlying mathematical models. This matters when a simulator is used to teach causal reasoning. If the internal physiology is opaque, learners can observe an output without being able to inspect why it occurred. If numerical failure is hidden, an apparent physiologic response may actually reflect a software artifact.

At the opposite end of the spectrum, sophisticated virtual-patient and digital-twin models increasingly use patient data to calibrate computational physiology and predict responses to ventilator changes. These systems are scientifically important, but they solve a different problem and generally require patient-specific data, model-identification pipelines, imaging, optimization, or other infrastructure unsuitable for lightweight educational deployment.

We therefore designed a mechanistic lung simulator with a deliberately narrower goal: make heterogeneous lung mechanics, recruitment, ventilator interaction, and numerical behavior explicit enough to inspect, test, and teach in a standard browser. The central contribution is architectural rather than rhetorical. The simulator does not claim to be the first web ventilator simulator, the first compartment model, or a clinical digital twin. Instead, it combines features that are usually separated across educational software and computational physiology research: an inspectable multi-compartment mechanics model, explicit recruitment state, reproducible initialization, strict separation of phenotype from scenario, explicit solver failure semantics, machine-readable numerical diagnostics, automated verification, and responsive mobile deployment.

2. Relationship to prior work and novelty

The contribution is best understood by separating three neighboring literatures: educational software simulators, physical/high-fidelity simulation, and patient-specific computational models.

Domain	Representative prior work	Primary strength	Gap addressed here
Web/software ventilation simulation	Takeuchi et al.; Tallo et al.; VentSim/Safadi et al.	Low-barrier access; interactive ventilator learning; educational feasibility	Adds inspectable compartment mechanics, explicit recruitment state, software-verification discipline, failure taxonomy, and machine-readable solver diagnostics.
Physical and high-fidelity simulation	ASL 5000 and mannequin-based curricula	Hands-on interaction with real ventilators; psychomotor and team learning	Targets the complementary cognitive/mechanistic domain in a browser without dedicated hardware.
Virtual patients / digital twins	Zhou et al.; Ang et al.; Weaver et al.; Joy et al.; Dunphy et al.	Patient-specific prediction, optimization, and research-grade physiologic inference	Avoids patient-specific claims; prioritizes transparent educational experimentation and reproducible software behavior.

This distinction is important because novelty in simulation software can be overstated if novelty is equated with “web-based,” “multi-compartment,” or “digital twin.” All three concepts have substantial prior literature. The defensible novelty claim is the integration of transparent mechanistic modeling with explicit software contracts and reproducible numerical verification in an educational, browser-native artifact.

3. Methods

3.1 Design principles

The simulator was designed around four principles: (1) physiologic assumptions should be visible in code and documentation; (2) clinical scenario choices should not be silently embedded inside phenotype definitions; (3) numerical failure should be classified and surfaced rather than

converted into plausible-looking output; and (4) the artifact should run on ordinary desktop and mobile browsers without dedicated simulation hardware.

3.2 Phenotype and scenario separation

The phenotype object contains intrinsic mechanical properties: compartment fractions, branch resistances, capacities, elastic stiffness, central airway resistance, airway opening pressure, perfusion fractions, and dead-space fractions. Scenario state contains the initial PEEP, initial recruitment state, ventilator mode, FiO₂, respiratory rate, tidal volume or inspiratory pressure, inspiratory timing, pause, and other control parameters. Missing initial PEEP or recruitment state is rejected explicitly rather than inferred. The four included phenotypes are labeled Reference, Low-recruitability, Moderate-recruitability, and High-recruitability; they are mechanical constructs and are not mapped to Berlin ARDS severity categories.

3.3 Three-compartment mechanics

Each tissue compartment has a recruitment-dependent available capacity and a finite-capacity exponential pressure-volume relationship. In the open regime, compartment volume approaches but cannot exceed the recruitment-dependent maximum volume. Closed compartments remain exactly closed. The lower-bound regime prevents negative compartment volume. Airway pressure is coupled to a lumped central resistance and per-compartment branch resistances, allowing the model to represent both elastic and resistive pressure components while conserving total flow.

For each compartment, the elastic relation is represented conceptually as $V = V_{\max}(r)[1 - \exp(-(P - AOP)/K)]$, where $V_{\max}(r)$ is recruitment-dependent available capacity, AOP is airway opening pressure, and K is elastic stiffness. The corresponding analytic derivative $dP/dV = K/(V_{\max} - V)$ is used directly by the Newton solver.

3.4 Recruitment and state transitions

Recruitment is represented as an explicit state variable rather than an implicit consequence of the phenotype label. This allows identical tissue parameters to be initialized under different histories and prevents hidden clinical choices from being baked into presets. Recruitment and derecruitment are constrained by capacity and projection rules so that a state transition cannot commit an impossible compartment volume.

3.5 Numerical solver and failure semantics

The nonlinear mechanics step solves compartment volume changes and branch pressure simultaneously. Convergence is dimensionlessly scaled so that residual tolerance is not weakened simply because a compartment has a large nominal volume. If the constrained nonlinear solve does not converge, a direction-aware feasibility classifier evaluates whether the requested flow is structurally impossible for the current state. Inspiratory requests are bounded by remaining available capacity; expiratory requests are bounded by removable gas volume. Requests that violate these necessary conditions are labeled `INFEASIBLE_BOUNDARY`. Other failed solves are labeled `SOLVER_NONCONVERGENCE`.

Failure semantics are transactional. A failed mechanics step does not advance simulation time, commit a partially solved volume, or leave controller/tracker state advanced. Version 0.4.5 extends rollback handling to the surrounding simulation pipeline so that higher-level state remains consistent with the mechanics state after a rejected step.

3.6 Ventilator controls and browser deployment

The browser interface exposes volume-control and pressure-control ventilation, FiO₂, PEEP, respiratory rate, tidal volume or inspiratory pressure, flow/timing parameters, and the four mechanical phenotypes. Version 0.4.5 introduces responsive phone, tablet, and desktop layouts; larger touch targets; separate pressure, flow, and volume plots; cancellation of long simulations; inline error reporting; and JSON export. Simulations execute in a Web Worker so longer runs do not block the main interface. Scenario validation rejects volume-control settings in which inspiratory time plus pause would leave no time for expiration.

3.7 Computational verification

The verification suite is deterministic and executable from the package root. It tests model contracts rather than visual plausibility alone. The v0.4.5 suite contains 143 assertions. Major verification domains include initialization, analytic Jacobian agreement, small-signal RC behavior, central and branch flow conservation, pressure consistency, low-resistance limiting behavior, recruitment projection, closed-compartment invariants, multi-breath stability, timestep convergence, deterministic results, controller behavior, gas-exchange toggles and metrics, scenario validation, rollback on failed steps, boundary classification in both flow directions, and per-step solver instrumentation.

4. Results

4.1 Automated verification

All 143 automated assertions passed, with zero failures, in the v0.4.5 release. This result supports internal computational consistency within the tested parameter domain. It does not establish clinical validity, cohort calibration, educational efficacy, or safety for patient care.

Verification domain	What is tested	Why it matters
Initialization and state contracts	Explicit PEEP/recruitment input; closed compartments; lower bounds	Prevents hidden defaults and impossible starting states.
Analytic mechanics	Jacobian agreement; RC behavior; pressure/flow conservation	Tests whether the solver implements the stated mechanics.
Recruitment and capacity	Projection, finite capacity, compartment invariants	Prevents recruitment transitions from creating invalid volumes.
Dynamic simulation	VC/PC multi-breath runs; timestep convergence; deterministic execution	Tests stability across time and controller modes.
Failure handling	Rollback; INFEASIBLE_BOUNDARY vs SOLVER_NONCONVERGENCE; positive/negative flow cases	Separates structural impossibility from numerical failure and prevents corrupted state.
Instrumentation	Newton iterations, substeps, line-search halvings, residuals, active-set transitions	Makes numerical behavior inspectable rather than hidden.
Browser/scenario layer	Input validation, worker execution, cancellation, pipeline rollback	Ensures deployment logic does not undermine mechanics-state integrity.

4.2 What changed in v0.4.5

Version 0.4.5 is principally a deployment and interaction release. The core constitutive mechanics, recruitment equations, phenotype parameters, and gas-exchange module remain unchanged from the preceding verified core. The material additions are a responsive mobile/desktop interface, Web Worker execution, cancellation, clearer plotting, JSON export, stricter scenario validation, and rollback of surrounding controller/tracker state when a mechanics step fails. The automated test count increased from 127 in v0.4.4.1 to 143 in v0.4.5.

4.3 Interpretable failure as an educational output

An unusual feature of the simulator is that failure itself is treated as information. A user request can be rejected because the state has insufficient remaining capacity for the requested inspiratory flow, because insufficient gas volume exists for the requested expiratory removal, or because the numerical solver did not converge despite passing the fast feasibility checks. These conditions are surfaced separately. For education, this avoids a common failure mode of simulation software: silently returning a number that appears physiologic even when the underlying step was not mathematically admissible.

5. Discussion

5.1 Principal contribution

The principal contribution is a transparent and testable bridge between educational simulation and computational respiratory modeling. Educational web simulators have shown that browser-based tools can support mechanical ventilation learning. Patient-specific virtual-patient and digital-twin studies have shown that mechanistic models can reproduce and predict clinically relevant respiratory behavior when calibrated to measured data. This project deliberately does not compete with either category on its own terms. Instead, it borrows the accessibility of educational software and the explicitness of mechanistic modeling, then adds software-verification features that allow reviewers and learners to inspect when the model succeeds, when it fails, and why.

5.2 Why this is novel without claiming “first”

The published literature makes several broad “first” claims indefensible. Browser-based ventilator simulation predates this project. Multi-compartment respiratory models predate it. Recruitment modeling and patient-specific virtual lungs predate it. The more defensible novelty is the combination of features in a single educational artifact:

- An explicit three-compartment heterogeneous lung model in which recruitment state is visible and separable from phenotype.
- An architecture that distinguishes intrinsic tissue mechanics from scenario-specific clinical decisions, preventing hidden initialization choices.
- An analytic, instrumented nonlinear solver whose failure modes are exposed to the calling application.
- Direction-aware boundary classification that distinguishes impossible inspiration from impossible expiration and both from solver non-convergence.
- Transactional failure semantics that preserve mechanics, controller, and tracker state when a step is rejected.
- A deterministic acceptance suite that tests numerical contracts rather than relying on visually plausible waveforms.

- A responsive, browser-native deployment that runs on common desktop and mobile devices without dedicated hardware while retaining access to the mechanistic state.

This combination is the paper’s strongest novelty argument. It is specific, testable, and avoids overstating what the broader field has already accomplished.

5.3 Educational relevance

Mechanical ventilation education benefits from cognitive experimentation: changing PEEP, tidal volume, inspiratory pressure, respiratory rate, or timing and then observing how pressures, flows, volumes, recruitment, and gas exchange respond. Software simulators are well suited to this domain because they permit rapid repetition and can expose internal variables that are difficult or impossible to measure at the bedside. The current simulator extends that educational affordance by making internal model state and solver status inspectable, which may support teaching not only “what changed?” but also “what mechanism produced the change?”

However, educational value has not yet been measured for this specific simulator. Published evidence supporting virtual ventilation simulation in general should not be conflated with validation of this artifact. A future study should evaluate usability, expert face validity, knowledge acquisition, transfer to waveform interpretation, and whether explicit access to compartment mechanics improves causal reasoning compared with a conventional black-box simulator.

5.4 Relationship to digital twins

The term digital twin is intentionally not used for the present artifact. In contemporary respiratory modeling, digital twins and virtual patients are commonly calibrated to measured individual-patient data and used to predict responses to changes in support, estimate unobservable variables, test protocols, or optimize ventilation. The current simulator has no patient-specific calibration pipeline and its four phenotypes are didactic mechanical constructs. Calling it a digital twin would therefore blur an important methodological distinction. A future patient-specific layer could use the present mechanics engine as one component, but that would require calibration, validation against measured data, uncertainty quantification, and a separate clinical research program.

5.5 Limitations

- The phenotypes are prespecified mechanical constructs and are not derived from clinical ARDS cohorts or mapped to Berlin severity categories.
- The model has not been prospectively validated against patient waveforms, imaging, regional ventilation data, or clinical responses to ventilator changes.
- No expert face-validity study or learner-outcome study has been completed.
- The three-compartment representation deliberately trades anatomic resolution for interpretability and browser performance; it cannot represent the spatial detail of CT-based or high-dimensional lung models.
- The current release is not a patient-specific model and must not be used for bedside ventilator selection or clinical decision support.
- Responsive browser deployment has been engineered and tested in software, but formal physical-device usability testing should precede any claim of production-grade mobile deployment.

6. Conclusion

This work presents an open, browser-native mechanistic lung simulator whose main contribution is the integration of transparent heterogeneous lung mechanics with explicit numerical contracts, reproducible computational verification, and low-barrier deployment. The field already contains web ventilator simulators, high-fidelity training systems, multi-compartment lung models, and patient-specific digital twins. The novelty here lies in combining mechanistic inspectability with software-verification discipline in an educational artifact: the phenotype is separated from the scenario, recruitment state is explicit, infeasible boundaries are distinguished from solver non-convergence, failed steps roll back cleanly, solver behavior is instrumented, and the same engine is exposed through a responsive browser interface. The v0.4.5 release passes 143 automated assertions with zero failures within the tested domain. These results establish computational consistency, not clinical or educational validity. The next phase should test physiologic fidelity, expert face validity, usability across devices, and learner outcomes.

7. Availability and reproducibility

Simulator and manuscript site: <https://vent.defying-logic.com>

Source repository: <https://github.com/DrFillster/medical-vent-twin>

Version: v0.4.5. The release package includes the browser application, source, tests, and machine-readable verification artifacts. The automated suite is intended to be rerunnable from a clean package checkout.

8. Declarations

Ethics approval: Not applicable to the computational development report; no human participants or patient data were used in the reported verification work.

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Data and code availability: Source code and release artifacts are publicly available at the repository listed above.

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